TWINPULSE · DIGITAL TWIN CHALLENGE 2026

The twin lives the night first



A physiology-based digital twin that rehearses tonight to prevent night-time hypoglycaemia in Indian children with type 1 diabetes

21:00 → 07:00
500 nights rehearsed
before one is lived

Team TwinPulse

2:47 AM in Raichur

“Every night for three years I set an alarm to prick her finger. I never know which night is the dangerous one.”

Composite story, for illustration

55%

of severe lows in type 1 diabetes happen during sleep (DCCT)

5–6%

of deaths in people with T1D under 40 are linked to “dead-in-bed” syndrome

~1 hour

of warning is enough to treat a low with a glass of juice

India has the most children with type 1 diabetes, and the least protection at night

2.29 lakh

Indian children and teens with T1D: the most of any country (IDF Atlas 2021)

9.4 lakh

Indians of all ages with T1D (IDF Atlas 2025)

₹1.1 lakh+

a year for a CGM with low alarms (₹4–4.7k per 14-day sensor)

NPH

Cheaper human insulin many programmes supply peaks between 1 and 4 AM, when everyone is asleep

Families choose between ₹1 lakh a year and a 2 AM finger-prick alarm. Fear of lows pushes sugars high, which harms eyes and kidneys later. Clinics have no way to tell which nights are dangerous.

TwinPulse: learn the child, live tonight first, guard the real night

1

Learn the child

A personal glucose–insulin model is fitted to each child's own nights: insulin, meals, play, CGM or finger-pricks, plus the FHIR health record.

2

Live tonight first

At 9 PM the twin simulates tonight 500 times. The doctor sees who is at risk, when and why, and rehearses a fix on the twin before prescribing it.

3

Guard the real night

For CGM users, a model trained on real patients warns about an hour before a low and calls the parent in their own language.

Decision support for clinicians. The doctor approves every change; families receive only approved actions.

Fusing the health record with wearable streams

Static: health record (FHIR R4)

- Age, weight, HbA1c, haemoglobin
- Insulin regimen and daily dose
- Past severe lows, hypo awareness
- HLA genotype, islet autoantibodies

+

Dynamic: wearables

- CGM every 5 min, or finger-pricks
- Insulin doses and meals
- Steps and heart rate from a band
- Sleep window

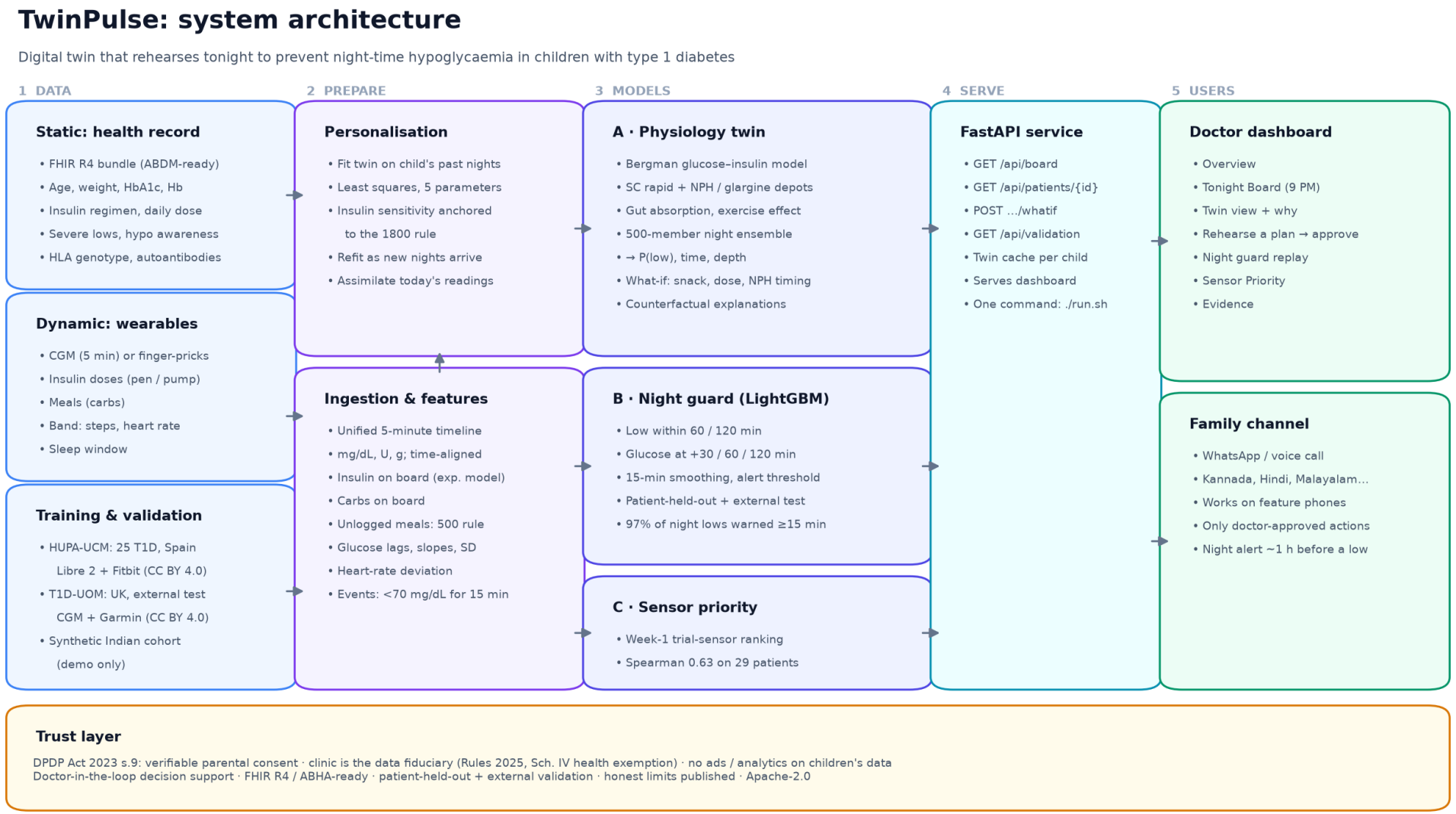
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Twin output

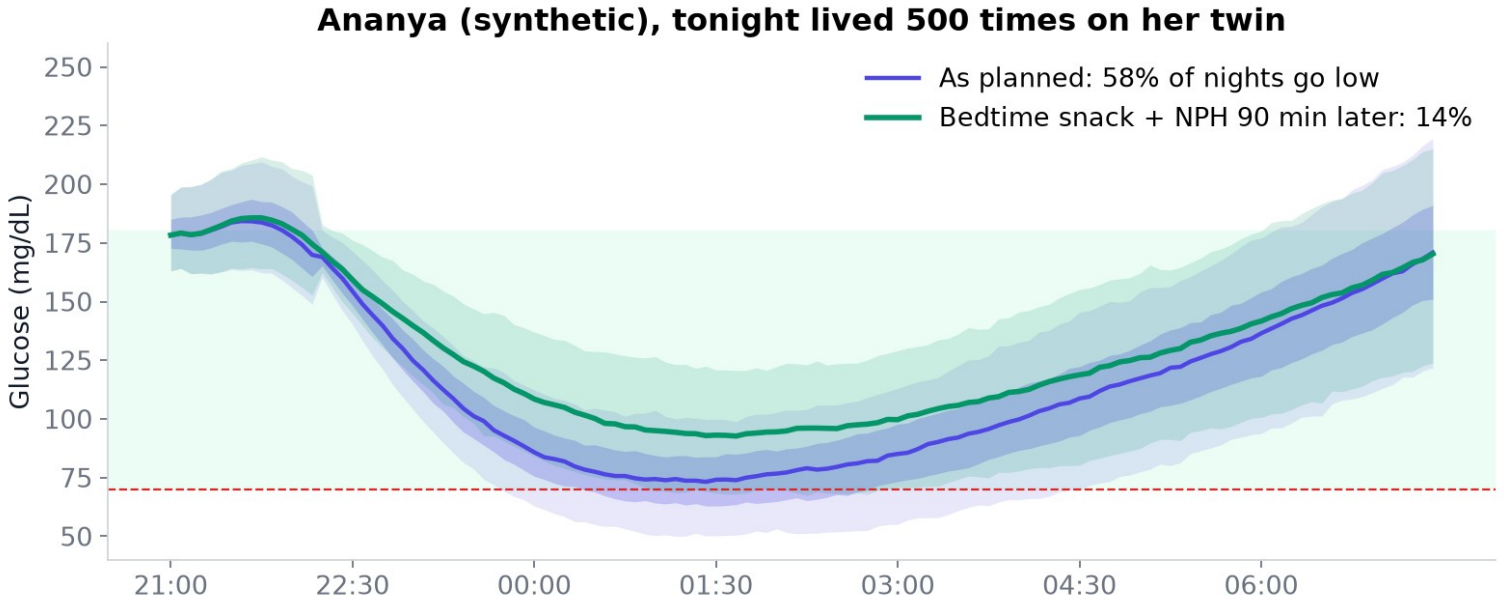
- Tonight's risk, time and depth of a low
- Factor-by-factor explanation
- Rehearsed plan, doctor-approved
- Real-time alert to the parent

Dataset	What it contains	Role
HUPA-UCM (Spain, CC BY 4.0)	25 people with T1D: Libre 2 CGM, Fitbit heart rate and steps, insulin, carbs	Training + patient-held-out validation
T1D-UOM (UK, CC BY 4.0)	T1D, 3 months each: CGM, pump and injected insulin, food, Garmin activity, sleep	External test
Synthetic Indian cohort	12 children, 14 days, Indian routines, NPH/premixed/glargine, FHIR records	Dashboard demo only

Architecture



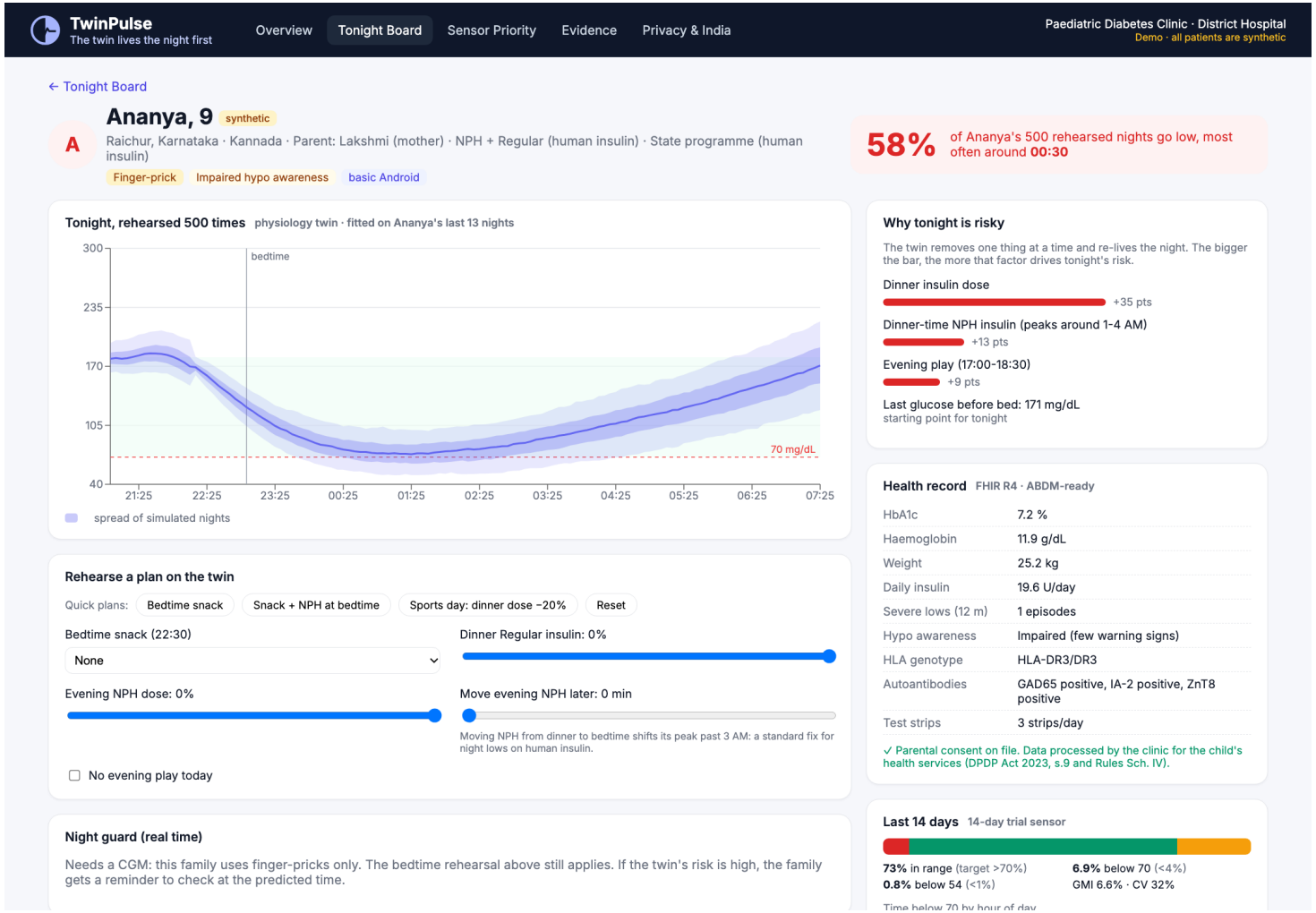
The physiology twin: a personal model of each child



Synthetic patient · the method, not an accuracy claim

- Bergman glucose–insulin model with rapid and long-acting (NPH or glargine) insulin depots, gut absorption and a 4-hour exercise effect
- Fitted on the child's own past nights; insulin sensitivity anchored to the clinical 1800 rule
- Unlogged meals inferred from mealtime insulin (500 rule)
- Assimilates today's readings, then runs 23:00–07:30 as a 500-member ensemble

Rehearse the fix on the twin before prescribing it

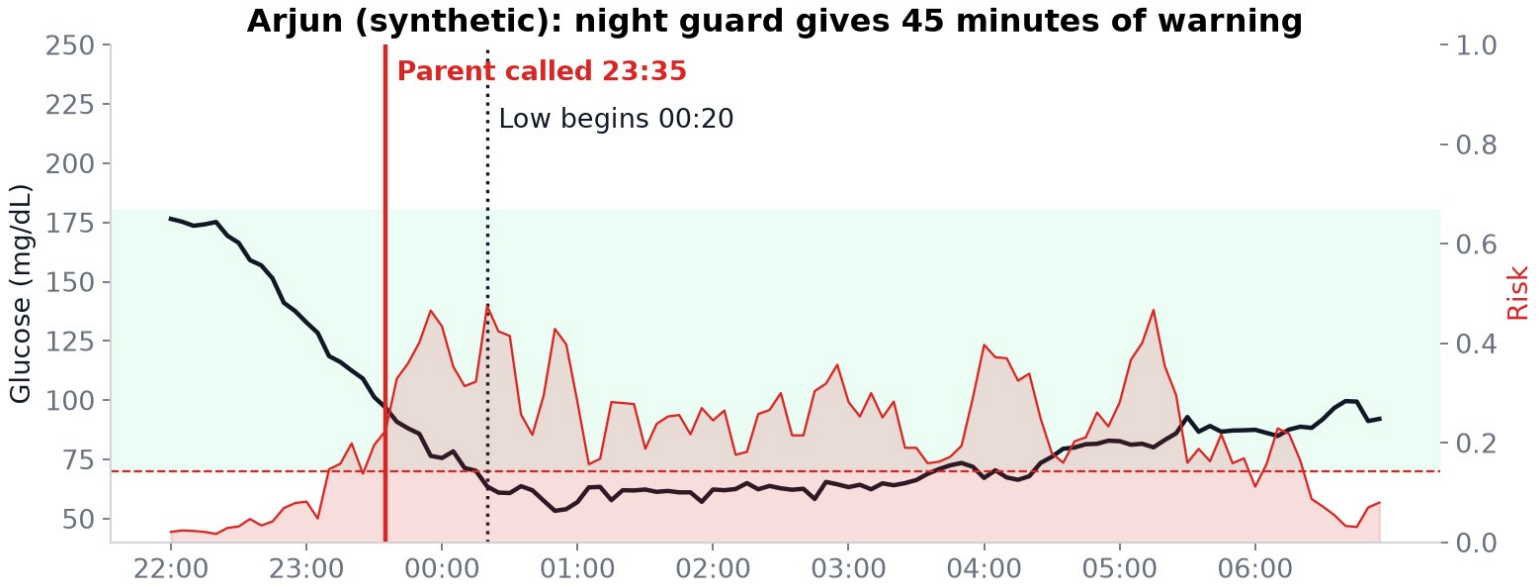


58% → 14%

Ananya's simulated risk of a night low, before and after: 15 g bedtime snack + NPH moved 90 min later

- Bedtime snack (15 or 30 g)
- Smaller dinner insulin dose after a sports day
- Move NPH from dinner to bedtime
- Explanations: the twin removes one factor at a time and re-lives the night

Night guard: wake the parent before the low, not after



LightGBM on glucose history, insulin and carbs on board, steps and heart rate; 15-minute smoothing. Alert goes as a voice call in the family's language.

97%

of 315 real night-time lows warned at least 15 min ahead (unseen patients)

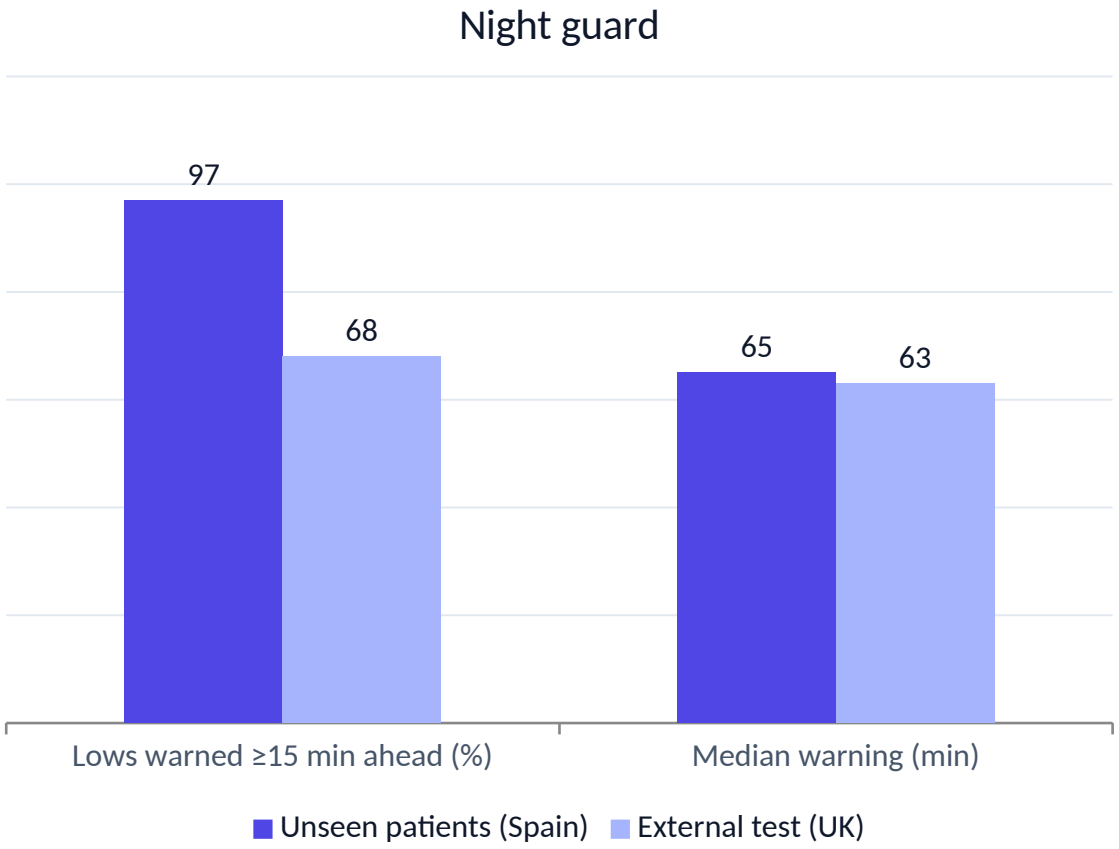
65 min

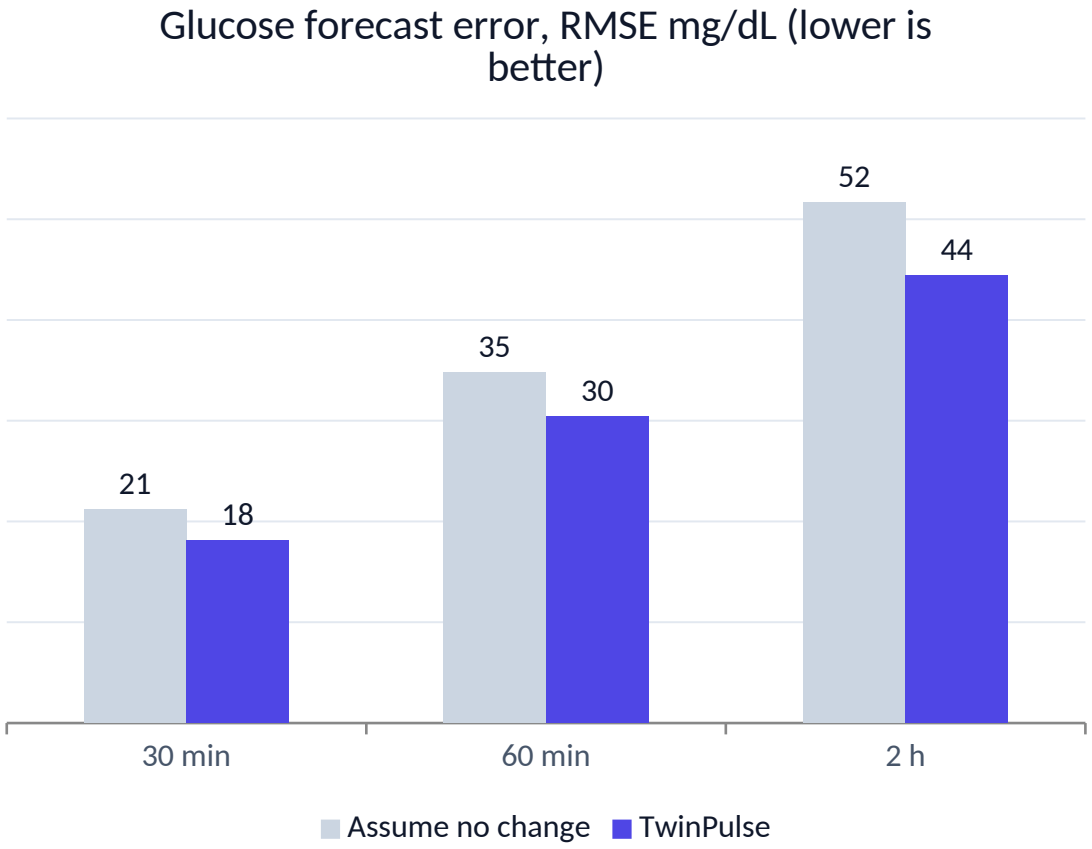
median warning

0.55

false alarms per night

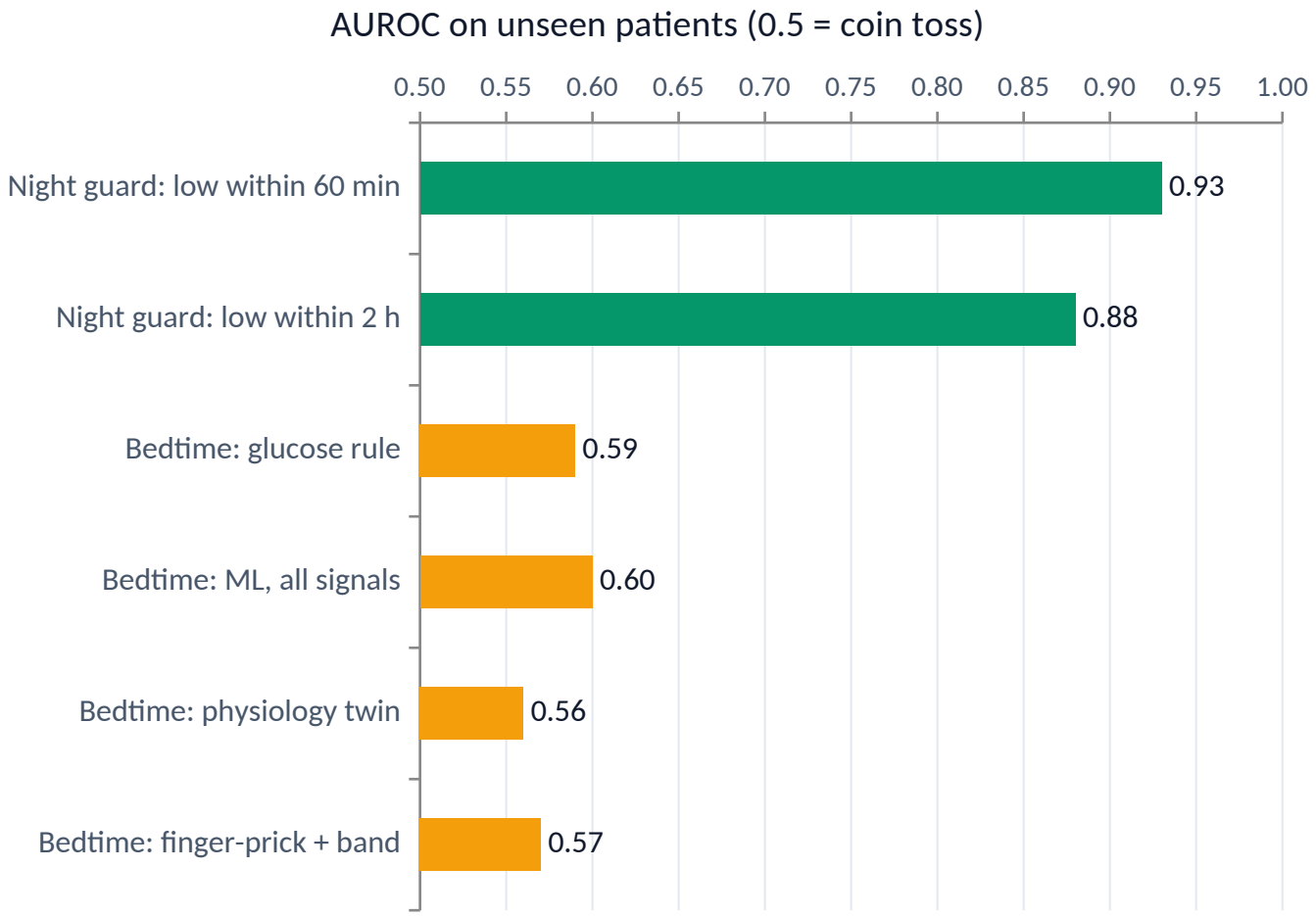
Results on real patients the model never saw





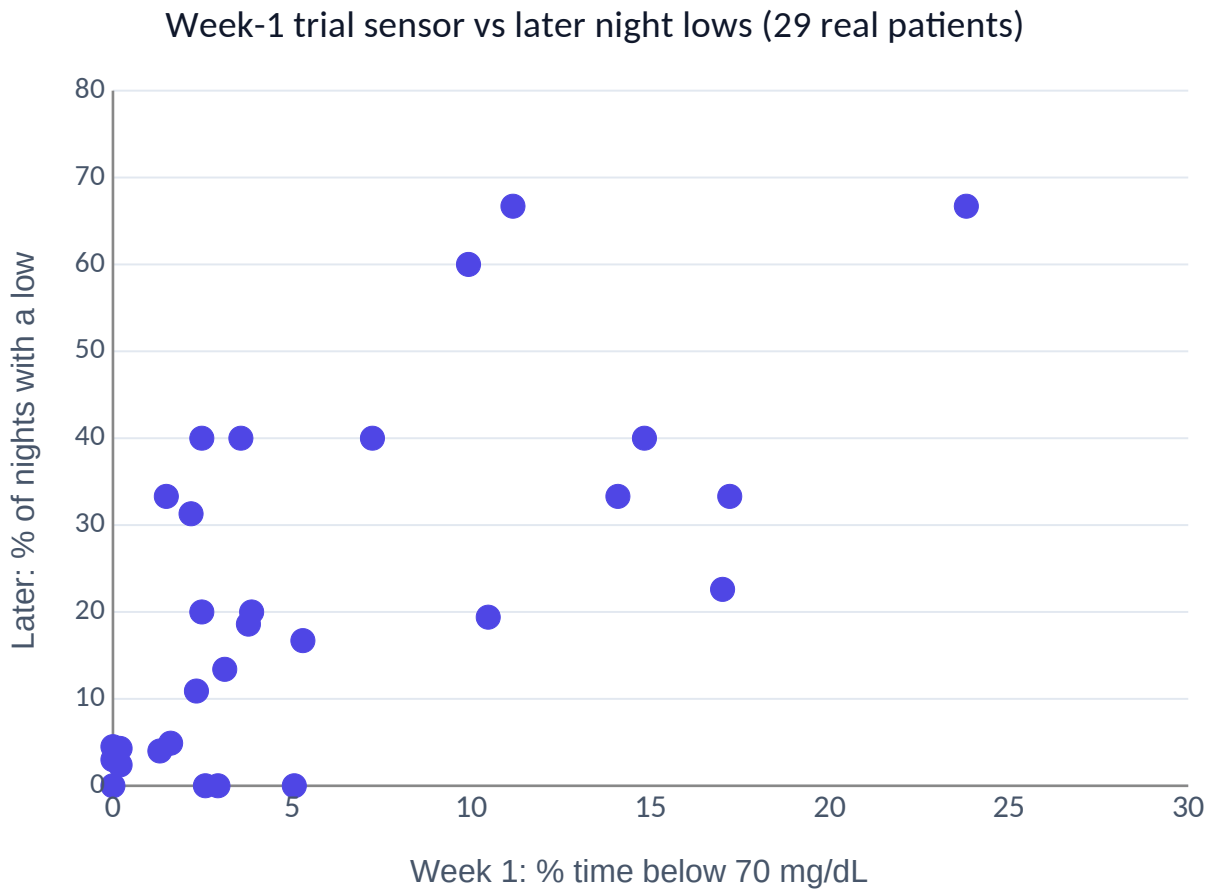
Night-guard AUROC: 0.93 for a low within 60 min, 0.88 within 2 h. Clarke error grid A+B at 30 min: 97.6%. Patients split into 5 groups, so every score is on people the model never saw.

What did not work, and what we did about it



- Predicting the whole night from bedtime alone was close to chance (0.55–0.63) on 1046 real nights, whatever the method
- So the bedtime rehearsal is used to compare plans, not to promise a night is safe
- Warnings rely on the in-night guard
- Insulin, meal, step and heart-rate features add only ~0.01 AUROC over glucose alone in these datasets

No CGM for everyone? Give it to the children who need it most



42%

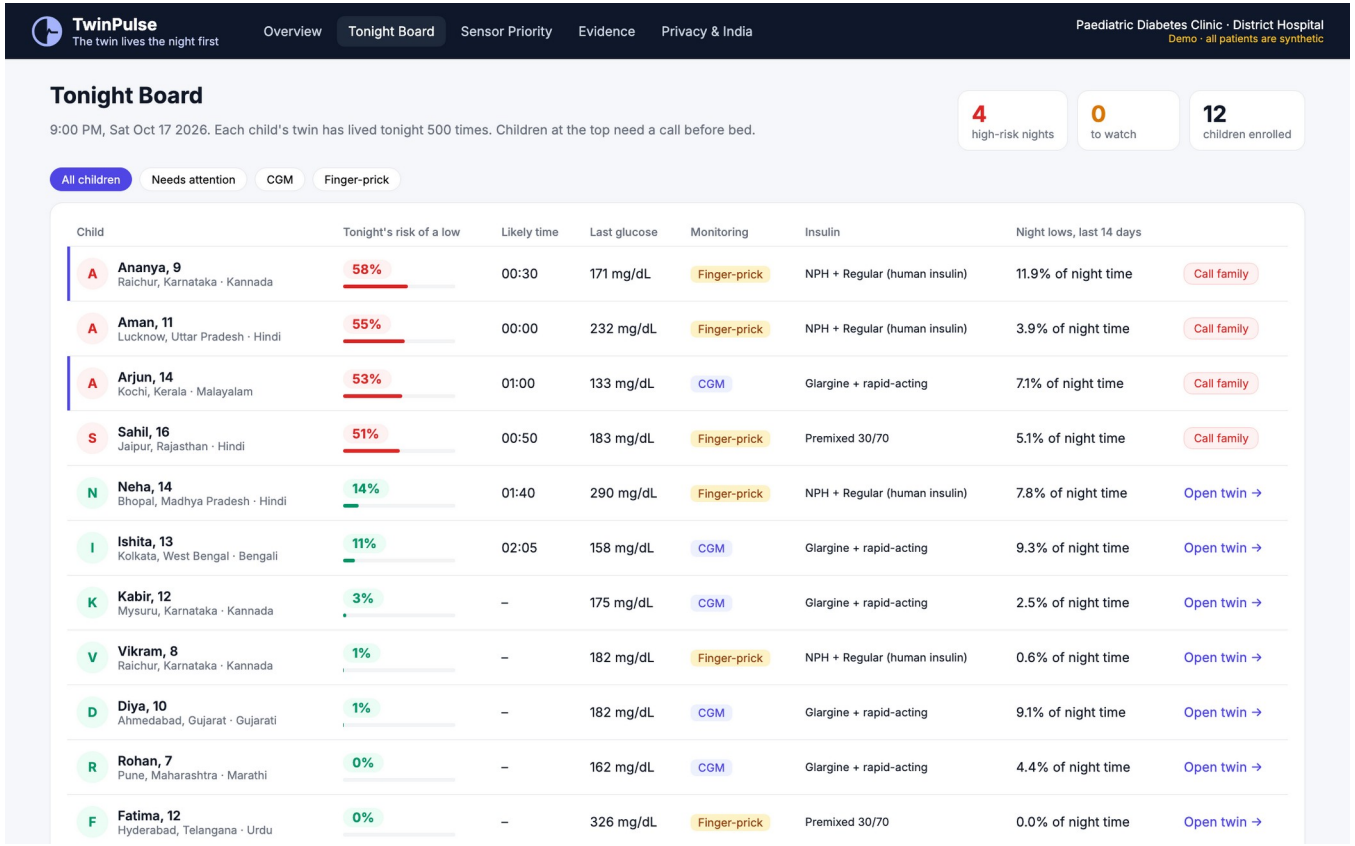
of later nights had a low: top third by week-1 risk

10%

bottom third

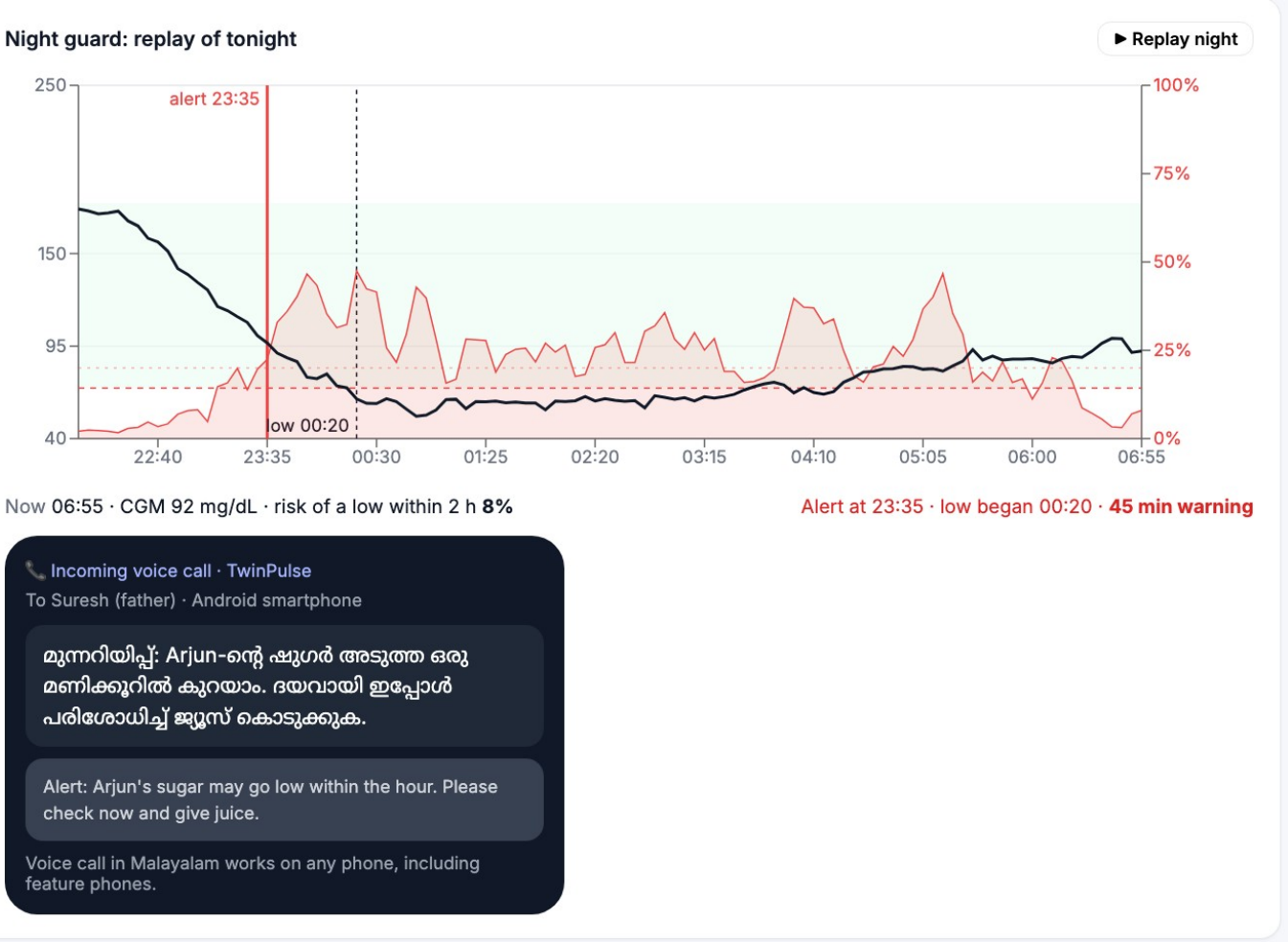
Spearman 0.63. One ₹4–5k trial sensor per child, instead of ₹1.1 lakh a year for everyone, points a state programme's scarce CGMs at the children most at risk. The Sensor Priority screen does exactly this.

The doctor's dashboard



- Tonight Board at 9 PM: every child ranked by risk; call the red rows
- Twin view: rehearsed night, why it is risky, FHIR record, 14-day time in range
- Rehearse, approve, send to family
- Night guard replay
- Sensor Priority and Evidence pages

Built for Indian families and clinics



1. **Language**

   Alerts in Kannada, Hindi, Malayalam… by voice call that works on feature phones

2. **Insulin reality**

   Models human NPH/Regular and premixed insulin, not only analogues

3. **Routines**

   Late dinners, evening cricket, school tiffin

4. **ABDM-ready**

   FHIR R4 records with room for ABHA linkage

Privacy, safety and regulation

§

Children's data (DPDP Act 2023)

Section 9: verifiable parental consent; no tracking or behavioural monitoring of children. The DPDP Rules 2025 (Fourth Schedule) exempt clinics and healthcare professionals for health services to the child, so TwinPulse is deployed by the clinic as data fiduciary, with consent recorded anyway.

✓

Doctor in the loop

The twin only shows simulated outcomes. The doctor approves every change; families receive only approved actions. No advertising or analytics on children's data.

→

Path to the clinic

Decision support, not a diagnosis. A real deployment would follow CDSCO's medical device software pathway after a supervised clinical pilot.

Impact and how it scales

~1 hour

of warning before a night-time low for CGM families, as a voice call

4.3×

more night lows in the top third flagged by a trial sensor, so scarce CGMs go further

₹0

licence cost: open source (Apache-2.0), deployable by state programmes and NGOs

Rollout path

- Pilot in one paediatric diabetes clinic with families already on a state T1D programme
- Recalibrate and validate on Indian children's data; measure night lows, parent sleep and false alarms
- Extend the same twin to adults with T2D on insulin or sulfonylureas

Limitations we know about

Training population

Models trained on adults in Spain and tested on adults in the UK, not Indian children. An Indian pilot is the next step.

Synthetic demo

The 12 Indian children in the dashboard are synthetic. They show the method; they are never used for accuracy numbers.

Bedtime prediction

Bedtime-only risk is weak on real data. We use it to compare plans, not to promise safety.

Translations and regulation

Translations are drafts for native-speaker review. Not a medical device yet.

Let the twin face the night first



So a mother can sleep, and a child wakes up safe

Team TwinPulse

Code, data sources and results: see the GitHub README